Validation Criterion Determination Module (VCDM)

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Governed Space

Validation Criterion Determination

Canonical Definition

Validation Criterion Determination Module (VCDM) defines the governance framework for evaluating qualified Validation Results against individual applicable Validation Criteria and formally determining whether each criterion is **Satisfied, Conditionally Satisfied, Not Satisfied, or Not Determinable** within the validated object, scope, depth, thresholds, evidence conditions, source conditions, execution limitations, and methodological boundaries established by VALIDOS®.

It establishes the first formal object-level conclusion layer within Validation Determination.

Operational Role

VCDM serves as the second internal governance space of the Validation Determination Standard.

The preceding module establishes:

VDBRQM — Which validation results may legitimately participate in determination?

VCDM asks:

What do those qualified results demonstrate relative to each individual Validation Criterion?

The governed progression is:

Qualified Determination Basis → Criterion Activation → Result-to-Criterion Mapping → Threshold Assessment → Condition Assessment → Uncertainty Assessment → Criterion Determination → Criterion Determination Record

VCDM does not yet integrate all criteria into an overall Validation Determination.

It determines each criterion independently before cross-criterion interpretation begins.

Module Operational Space

VCDM governs:

- criterion activation,
- criterion applicability,
- criterion-level determination,
- result-to-criterion mapping,
- criterion evidence basis,
- criterion threshold assessment,
- criterion acceptance conditions,
- criterion rejection conditions,
- criterion satisfaction,
- conditional criterion satisfaction,
- criterion non-satisfaction,
- criterion non-determinability,
- mandatory criterion status,
- critical criterion status,
- criterion dependencies,
- criterion prerequisites,
- criterion exclusions,
- criterion uncertainty,
- criterion limitations,
- criterion scope,
- criterion depth,
- criterion determination confidence,
- criterion determination conflict signals,
- criterion determination versioning,
- criterion reassessment,
- and criterion-determination traceability.

Module Function

VCDM applies after the Validation Determination Basis has been established and material results have been qualified.

Its function is to prevent:

- raw results being interpreted without explicit criterion linkage,
- criterion satisfaction being assumed from general system performance,
- missing evidence being converted into criterion failure,
- missing evidence being converted into criterion success,
- thresholds being changed after results are known,
- conditional findings being represented as unconditional,
- one strong result overriding multiple contradictory results without governance,
- mandatory and critical criteria losing their significance,
- criterion scope expanding beyond validated scope,
- criterion outcomes being detached from Validation Depth,
- and overall determination being constructed before individual criterion outcomes are explicitly established.

Criterion Activation

Not every Validation Criterion is necessarily applicable to every Validation Object or Validation Execution.

VCDM therefore begins by establishing which criteria are active for the determination.

A criterion may be:

Applicable

Conditionally Applicable

Not Applicable

or:

Applicability Undetermined

Criterion applicability should derive from the governed Validation Criteria Framework rather than being improvised during determination.

Criterion Identity

Every active criterion should be sufficiently identifiable.

A Criterion Determination Record may include:

- Criterion Identifier,
- criterion version,
- criterion definition,
- criterion type,
- mandatory status,
- critical status,

- threshold,
- acceptance conditions,
- rejection conditions,
- scope,
- dependencies,
- prerequisites,
- and applicable Validation Depth.

Criterion-to-Result Mapping

VCDM maps qualified Validation Results to the criterion they are expected to inform.

The basic relationship is:

Validation Criterion → Qualified Results → Criterion Determination

Where multiple results contribute, their distinct roles should remain visible.

A result may function as:

- primary support,
- corroborating support,
- contrary support,
- contextual support,
- threshold evidence,
- or another defined contribution.

Result Relevance to Criterion

A result may be valid for determination generally while still not materially inform a specific criterion.

VCDM therefore preserves criterion-specific result relevance.

Determination Eligible Result ≠ Relevant to Every Criterion

Criterion Evidence Basis

Each criterion should have an identifiable evidence basis assembled from qualified results and their associated evidence and source conditions.

This basis should allow a reviewer to understand:

- what information supports satisfaction,
- what information supports non-satisfaction,
- what uncertainty remains,
- and what limitations apply.

Threshold-Based Criterion Determination

Some criteria depend upon explicit thresholds.

Examples may include:

- maximum error rate,
- minimum reliability level,
- response-time boundary,
- safety limit,
- confidence threshold,
- tolerance range,
- or another measurable condition.

VCDM governs whether the qualified results demonstrate performance relative to the approved threshold.

Threshold Integrity

Thresholds must remain those established through prior governance.

VALIDOS® establishes:

Thresholds must not be moved after results are known merely to produce a preferred validation outcome.

Where a threshold legitimately changes, the change should trigger criterion reassessment rather than silent reinterpretation.

Non-Threshold Criteria

Not every criterion can be reduced to a numerical threshold.

Some criteria may depend upon:

- documented existence,
- qualitative conditions,
- structural requirements,
- prohibited behaviors,
- procedural requirements,
- contextual acceptability,
- or expert interpretation.

VCDM supports criterion determination through appropriate governed logic rather than forcing all validation into numeric scoring.

Acceptance Conditions

A criterion may define conditions under which satisfaction can be established.

These may include:

- threshold achievement,
- required evidence combination,
- independent corroboration,
- absence of exclusion conditions,
- required operational behavior,
- required documentation,
- or another criterion-specific condition.

Acceptance conditions should be explicit.

Rejection Conditions

Some criteria may establish explicit rejection or non-satisfaction conditions.

For example:

Any critical safety override failure → Criterion Not Satisfied

Rejection conditions may be binary even where other parts of the criterion perform strongly.

Criterion Satisfied

A criterion receives **Criterion Satisfied** where the qualified Determination Basis sufficiently demonstrates that the criterion's required conditions have been met within the applicable validation boundaries.

A satisfied determination should remain bounded to:

- object version,
- scope,
- population,
- environment,
- operational state,
- time,
- Validation Depth,
- and applicable conditions.

Criterion Conditionally Satisfied

A criterion receives **Criterion Conditionally Satisfied** where the required condition is demonstrated only subject to explicit limitations or restrictions.

Examples include:

- satisfied only within a defined operating range,
- satisfied only for a specified population,
- satisfied only while a control remains active,
- satisfied only under a specific configuration,
- satisfied only with continuing monitoring,
- or satisfied only for a defined period.

Conditionality is part of the criterion outcome.

Criterion Not Satisfied

A criterion receives **Criterion Not Satisfied** where the qualified Determination Basis sufficiently demonstrates that one or more required criterion conditions have not been met.

This requires affirmative governed basis for non-satisfaction.

VALIDOS® establishes:

Criterion Not Satisfied must not be inferred merely from insufficient evidence.

Criterion Not Determinable

A criterion receives **Criterion Not Determinable** where the available governed validation record cannot legitimately establish either satisfaction or non-satisfaction.

Possible causes include:

- insufficient results,
- conflicting results,
- excessive uncertainty,
- invalid execution portion,
- unresolved evidence limitation,
- unresolved source limitation,
- insufficient Validation Depth,
- missing required scenario,
- or another methodological gap.

Not Determinable Is Methodologically Valid

VALIDOS® recognizes:

Not Determinable is a valid conclusion where certainty is not supported.

This protects the architecture from forcing every criterion into Pass or Fail.

Criterion Failure and Validation Failure Are Different

A criterion may be Not Satisfied.

That does not automatically determine the overall Validation Determination unless the criterion's governance significance requires it.

For example:

a supporting criterion may fail while mandatory criteria remain satisfied.

The overall effect belongs to later cross-criterion integration.

Mandatory Criterion

A **Mandatory Criterion** is a criterion whose satisfaction is required for a defined positive overall Validation Determination.

Where a mandatory criterion is Not Satisfied, the later overall determination may be constrained according to the approved criteria framework.

VCDM records mandatory status.

It does not invent its consequence after observing results.

Critical Criterion

A **Critical Criterion** represents a criterion whose failure or uncertainty may carry disproportionate methodological or operational significance.

Critical status may require stronger:

- evidence,
- Validation Depth,
- source reliability,
- execution confidence,
- or determination treatment.

Critical status should be established prior to final result interpretation where possible.

Mandatory and Critical Are Different

A criterion can be:

- Mandatory but not Critical,
- Critical but not Mandatory,
- both,
- or neither.

For example:

A critical criterion may trigger additional validation if Not Determinable, while a mandatory criterion may simply be required for positive determination.

VCDM preserves this distinction.

Criterion Dependencies

One criterion may depend upon another.

For example:

Criterion B can only be meaningfully determined if Criterion A is Satisfied.

VCDM governs such dependency where established by the Validation Criteria Framework.

Criterion Prerequisite Failure

Where a prerequisite criterion is not satisfied or not determinable, dependent criterion determination may require:

- suspension,
- conditional determination,
- non-determinability,
- or another governed response.

The dependent criterion should not be interpreted as though the prerequisite existed.

Criterion Exclusion Condition

Some criteria may include exclusion conditions that automatically prevent satisfaction.

For example:

If prohibited behavior X occurs under defined conditions, criterion cannot be classified as Satisfied.

VCDM preserves such logic.

Criterion Scope

Each criterion outcome applies only to the scope supported by its Determination Basis.

Criterion Scope may be narrower than the overall Validation Scope.

For example:

A criterion may be determined only for:

- one population,
- one subsystem,
- one operating state,
- one region,
- one time period,
- or one configuration.

Criterion Depth

A criterion outcome should preserve the Validation Depth actually achieved.

A criterion may be:

Satisfied at Preliminary Depth

but not yet support:

High-Assurance Satisfaction

if the deeper methodological requirement was not met.

Criterion Determination Strength

Where useful, VCDM may distinguish the methodological strength supporting a criterion outcome.

This may consider:

- result quality,
- evidence fitness,
- source reliability,
- execution conformity,

- corroboration,
- uncertainty,
- and Validation Depth.

Determination strength should not replace the criterion outcome itself.

Criterion Confidence

Some implementations may use a confidence representation.

Where used, confidence must remain distinct from criterion status.

For example:

Criterion Satisfied with moderate confidence

is different from:

Criterion Conditionally Satisfied

Confidence concerns epistemic strength.

Conditionality concerns applicability boundaries.

Criterion Uncertainty

Uncertainty may arise from:

- measurement,
- sampling,
- methodology,
- evidence,
- sources,
- execution,
- object variability,
- or conflicting results.

VCDM asks whether uncertainty is:

- immaterial,
- tolerable,
- material,
- or determination-blocking for the criterion.

Uncertainty-Compatible Satisfaction

A criterion may still be Satisfied where uncertainty remains but does not materially threaten the conclusion under the governed Validation Depth and threshold logic.

Perfect certainty is not required.

Uncertainty-Driven Conditionality

Where uncertainty restricts the conditions under which satisfaction can be defended, the outcome may become:

Criterion Conditionally Satisfied

Uncertainty-Driven Non-Determinability

Where uncertainty is large enough that satisfaction and non-satisfaction cannot be distinguished, the correct outcome becomes:

Criterion Not Determinable

Criterion Conflict Signal

VCDM may identify that qualified results support materially different criterion interpretations.

At this stage, the conflict is preserved as a **Criterion Conflict Signal.**

Detailed conflict integration belongs to the next VDTS module.

Criterion Determination Logic

For each criterion, VCDM should establish sufficient logic connecting:

Criterion Requirement → Qualified Results → Threshold / Condition Assessment → Uncertainty → Criterion Outcome

This prevents criterion determinations from becoming unexplained judgment labels.

Human Criterion Determination

Where human judgment is required, the determination should remain connected to:

- criterion logic,
- qualified results,
- applicable evidence,

- source conditions,
- limitations,
- and uncertainty.

The human evaluator should not be able to substitute personal preference for the governed criterion.

Automated Criterion Determination

Where criterion logic is sufficiently explicit, determination may be automated.

Automation may evaluate:

- thresholds,
- binary conditions,
- logical combinations,
- exclusions,
- mandatory dependencies,
- or structured decision rules.

The automated rule set should remain versioned and traceable.

AI-Assisted Criterion Determination

AI may assist with:

- mapping results to criteria,
- identifying contradictions,
- summarizing evidence,
- detecting missing information,
- or preparing determination rationale.

AI assistance must remain subordinate to the governed criterion logic.

Criterion Determination Record

A formal record should preserve, where applicable:

- Criterion Identifier,
- Validation Object,
- criterion version,
- criterion requirement,
- scope,
- Validation Depth,
- threshold,
- acceptance conditions,

- rejection conditions,
- mandatory status,
- critical status,
- qualified result set,
- relevant evidence,
- source conditions,
- execution conditions,
- uncertainty,
- limitations,
- conflicts,
- determination logic,
- Criterion Determination,
- determination authority where applicable,
- date,
- and sufficient traceability.

Criterion Determination Versioning

If a criterion determination changes, the previous determination should remain historically visible.

Possible transitions include:

Satisfied → Conditionally Satisfied Satisfied → Not Determinable

Conditionally Satisfied → Satisfied

Not Determinable → Not Satisfied

or other governed changes.

Criterion Reassessment

Reassessment may be triggered by:

- new qualified results,
- corrected execution records,
- evidence reassessment,
- Source Reliability change,
- changed threshold,
- changed criterion interpretation,
- new uncertainty,
- newly discovered deviation,
- or object change.

Criterion Determination Suspension

A criterion determination may require temporary suspension where new information materially undermines confidence but reassessment is incomplete.

Suspension preserves methodological caution without prematurely creating a new final outcome.

Criterion Determination Withdrawal

A criterion determination may be withdrawn where the basis supporting it is materially destroyed.

This may occur because of:

- invalid result set,
- wrong object version,
- compromised critical evidence,
- source failure,
- invalid execution,
- or fundamental criterion error.

Minimum Implementation Framework

1. Activate Applicable Criteria

Identify the criteria eligible for determination.

2. Establish Criterion Requirements

Preserve:

- definition,
- threshold,
- acceptance conditions,
- rejection conditions,
- mandatory status,
- critical status,
- dependencies,
- and Validation Depth.

3. Map Qualified Results

Associate legitimate Validation Results with each criterion.

4. Establish Criterion Evidence Basis

Identify the evidentiary and source conditions supporting the result set.

5. Evaluate Thresholds and Conditions

Determine whether acceptance, rejection, exclusion, or prerequisite conditions are satisfied.

6. Integrate Criterion-Level Uncertainty

Assess whether uncertainty affects the outcome.

7. Establish Criterion Determination

Classify each criterion as:

- Criterion Satisfied,
- Criterion Conditionally Satisfied,
- Criterion Not Satisfied,
- or Criterion Not Determinable.

8. Preserve Conflict Signals

Record material result disagreement requiring later conflict governance.

9. Establish Scope and Limitations

Bound the criterion determination to the conditions actually demonstrated.

10. Preserve Lifecycle Traceability

Maintain:

- Criterion Identifier,
- criterion version,
- Validation Object reference,
- qualified results,
- evidence,
- sources,
- execution references,
- threshold assessment,
- acceptance conditions,
- rejection conditions,
- uncertainty,
- limitations,
- scope,
- depth,
- status,
- determination rationale,
- versions,
- reassessment triggers,
- and sufficient lifecycle continuity.

Governance Outputs

VCDM may produce:

- Active Validation Criteria Register,
- Criterion Applicability Record,
- Criterion Determination Basis,
- Result-to-Criterion Map,
- Criterion Evidence Basis,
- Criterion Threshold Assessment,
- Criterion Acceptance Assessment,
- Criterion Rejection Assessment,
- Criterion Satisfaction Record,
- Conditional Criterion Satisfaction Record,
- Criterion Non-Satisfaction Record,
- Criterion Not Determinable Record,
- Mandatory Criterion Record,
- Critical Criterion Record,
- Criterion Dependency Map,
- Criterion Prerequisite Record,
- Criterion Exclusion Record,
- Criterion Uncertainty Record,
- Criterion Scope Record,
- Criterion Depth Record,
- Criterion Conflict Signal,
- Criterion Determination,
- Criterion Determination Record,
- Criterion Determination Reassessment Trigger,
- Criterion Determination Change Record,
- Criterion Determination Suspension Record,
- and Criterion Determination Withdrawal Record.

Use Case 1 — Autonomous AI Safety

Scenario

An autonomous AI system has a criterion requiring successful human override within a defined maximum response time.

Application

Qualified execution results show:

- all normal-state override tests satisfy the threshold,
- one degraded-network test exceeds the threshold,
- and one critical emergency-state result is missing because execution failed.

Result

VCDM does not average the successful tests.

The criterion may become:

Criterion Not Determinable

or:

Criterion Not Satisfied

depending upon the exact approved criterion logic governing the degraded failure and missing critical condition.

Use Case 2 — Healthcare AI

Scenario

A diagnostic model meets required accuracy thresholds for the overall validated population but evidence for one subgroup supports only a narrower operating condition.

Application

VCDM keeps the subgroup limitation attached to the criterion.

Result

The criterion may be:

Criterion Conditionally Satisfied

rather than universally Satisfied.

Use Case 3 — Industrial System

Scenario

A safety criterion requires that system temperature never exceed an approved limit during a defined stress test.

All average measurements are well below the limit, but one legitimate qualified result exceeds it.

Application

VCDM applies the approved rejection condition rather than averaging all measurements.

Result

If the criterion defines any threshold exceedance as disqualifying, the criterion becomes:

Criterion Not Satisfied

despite favorable average performance.

Architectural Position

Validation Criterion Determination is the second internal governance space of the **Validation Determination Standard (VDTS).**

The progression now becomes:

Determination Basis & Result Qualification → Validation Criterion Determination

VDBRQM establishes:

Which results may legitimately participate?

VCDM establishes:

What does that qualified result set demonstrate about each individual Validation Criterion?

The next VDTS governance stage can then ask: **How should contradictions, uncertainty, conflicting criterion evidence, and unresolved interpretive conditions be governed before multiple criterion outcomes are integrated?**

The VDTS internal progression therefore continues:

Determination Basis & Result Qualification → Criterion Determination → Conflict & Uncertainty Governance → Cross-Criterion Integration → Overall Validation Determination

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Foundational Principle

A criterion should be declared satisfied only when the governed validation record demonstrates satisfaction of that criterion—not merely because the overall system performed well.

Canonical Closing Statement

Validation Criterion Determination Module establishes the criterion-level conclusion layer of VALIDOS® by governing how qualified Validation Results are interpreted against individual Validation Criteria, thresholds, acceptance and rejection conditions, mandatory and critical status, dependencies, scope, depth, evidence, sources, execution conditions, uncertainty, and limitations. VCDM ensures that every criterion receives the outcome the governed validation record can actually support—Satisfied, Conditionally Satisfied, Not Satisfied, or Not Determinable—before those outcomes are allowed to influence any broader Validation Determination.

Parent Resources

→ VALIDOS® Validation Governance Architecture — Validation Governance Layer

→ VALIDOS® Architecture Map

→ VALIDOS® Complete Standards & Modules Index

Internal Module Flow

→ Previous module: Validation Determination Basis & Result Qualification Module (VDBRQM)

→ Next module: Validation Determination Conflict & Uncertainty Governance Module (VDCUGM)

Related Documents

→ Validation Determination Standard

→ About Validation Determination Standard

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Canonical Source

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